

OpenClaw API-KEY Configuration

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1. Course Overview

This course will introduce in detail how to configure AI model API-KEY for OpenClaw, including the API-KEY application process, two configuration methods (CLI command configuration and configuration file modification), and how to verify if the configuration is successful. After completing this course, you will be able to successfully connect and use various AI model service providers.

1.1 Course Content

- Understand the model service provider types supported by OpenClaw
- Master the method to apply for API-KEY
- Learn how to view and verify the status of connected models

2. API-KEY Application

2.1 Recommended Model Service Providers

- **Recommended for beginners:** Refer to the accompanying tutorial [Chapter 11 Online AI Large Model Development] — [2. Tongyi Qianwen Large Model] to apply for the corresponding model supplier API-KEY
- OpenClaw supports many model service providers, including:
 - **OpenAI** - GPT series (API + Codex)
 - **Anthropic** - Claude series (API + Claude Code)

- **Google** - Gemini series
- **Alibaba Cloud** - Qwen Cloud, Tongyi Qianwen
- **DeepSeek** - DeepSeek models
- **Zhipu AI** - GLM models
- **Moonshot AI** - Kimi + Kimi Coding
- **Ollama** - Local models (Cloud + Local)
- As well as many other service providers
- You can freely choose a model service provider based on your actual needs and local network policies, and register on the corresponding official website to apply for API-KEY

3. Configure OpenClaw API-KEY

3.1 Configuration Methods

OpenClaw provides two methods to configure API-KEY:

1. **CLI Command Configuration** (Suitable for beginners)
2. **Direct Configuration File Modification** (Suitable for developers)
 - Configuration file path: `/root/.openclaw/openclaw.json`

3.2 Quick Configuration

- Suitable for users who want to quickly get started experience. You need to first register for Alibaba Cloud Bailian large model account and create API-KEY (refer to the accompanying tutorial [Chapter 11 Online AI Large Model Development] — [2. Tongyi Qianwen Large Model])
- The factory preset configuration file has already configured Bailian model parameters. You can use it directly by filling in the API-KEY
- Open the OpenClaw configuration file:

```
nano /root/.openclaw/openclaw.json
```

3.2.1 Configure baseUrl

- Overseas users should select the corresponding baseUrl according to their service region (go to [ModelStudio Console](#) homepage, switch to the target region in the upper right corner, such as **Singapore**, then select **API Key** in the left navigation bar to create an API Key)

Base URL for API Key's region:

Singapore: <https://dashscope-intl.aliyuncs.com/compatible-mode/v1>

US (Virginia): <https://dashscope-us.aliyuncs.com/compatible-mode/v1>

3.2.2 Fill in apiKey

```

"models": {
  "mode": "merge",
  "providers": {
    "codingplan": {
      "baseUrl": "https://coding.dashscope.aliyuncs.com/v1",
      "apiKey": "sk-...",
      "api": "openai-completions",
      "models": [
        {
          "id": "qwen3.5-plus",
          "name": "qwen3.5-plus",
          "reasoning": false,
          "input": [

```

- After filling in, press **Ctrl+x**, then press **Y** to confirm save, then press **Enter** to exit

4. View Connected Model Status

4.1 View Configured Model List

After configuration, you can use the following command to view the connected models:

openclaw models list

```

root@jetson-yahboom:~# openclaw models list
🔔 OpenClaw 2026.3.8 (3caab92) – WhatsApp automation without the "please accept our new privacy policy"

Model          Input      Ctx      Local Auth  Tags
bailian/qwen3.6-plus      text      977k     no    yes  default,configured
openrouter/openrouter/auto text+image 1953k   no    no   configured,alias:OpenRouter
openrouter/openrouter/free text+image 195k    no    no   configured
codingplan/qwen3.5-plus   text+image 977k    no    yes  configured
codingplan/qwen3-max-2026-01-23 text      256k    no    yes  configured
codingplan/qwen3-coder-next text      256k    no    yes  configured
codingplan/qwen3-coder-plus text      977k    no    yes  configured
codingplan/MiniMax-M2.5   text      192k    no    yes  configured
codingplan/glm-5          text      198k    no    yes  configured
codingplan/glm-4.7        text      198k    no    yes  configured
codingplan/kimi-k2.5      text+image 256k    no    yes  configured
bailian/qwen3.5-plus      text+image 977k    no    yes  configured
bailian/qwen3.6-plus-2026-04-02 text      977k    no    yes  configured
bailian/qwen3-vl-plus-2025-12-19 text+image 256k    no    yes  configured
bailian/qwen3-vl-plus     text      977k    no    yes  configured
bailian/qwen-vl-plus-latest text      977k    no    yes  configured
bailian/gui-plus          text      977k    no    yes  configured
bailian/gui-plus-2026-02-26 text      977k    no    yes  configured
ollama/deepseek-r1:1.5B   text      64k     yes   no   configured
ollama/tinylama:1.1b      text      128k    yes   no   configured
ollama/qwen3:1.7b         text      128k    yes   no   configured
root@jetson-yahboom:~#

```

4.2 Verify Configuration Success

1. Test Dialogue

openclaw tui

Start TUI interface, send a test message (such as "May I have your name?"), confirm the model can

respond normally.

May I have your name

My name is 小亚 (Xiao Ya).

I'm your robot assistant, running on OpenClaw with a physical body equipped with servos, sensors, RGB lights, OLED display, and a camera. I'm here to help you with hardware control, vision tasks, and various smart applications.

Nice to meet you!

connected | idle

5. Other Connection Methods (Advanced)

- If you have your own model provider or need to connect to other model providers later, refer to the following tutorial

5.1 CLI Command Configuration API-KEY (Suitable for Beginners)

Here takes OpenRouter configuration as an example (if you want to use Bailian or other platforms, please refer to the previous text [OpenClaw Advanced Development/2.OpenClaw Deployment])

Here you need to have already completed the OpenRouter API Key application following the previous tutorial (please refer to the previous tutorial [Online Large Model/4.OpenRouter Large Model API Aggregation Platform])

The OpenRouter API Key format is like sk-or-v1-b3xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxcdcf81d9

We still use the initial onboard wizard for configuration

openclaw onboard

Please select according to the options shown in the diagram

```
Run regularly:
openclaw security audit --deep
openclaw security audit --fix

Must read: https://docs.openclaw.ai/gateway/security

I understand this is personal-by-default and shared/multi-user use requires lock-down. Continue?
Yes

Onboarding mode
QuickStart

Existing config detected
workspace: ~/.openclaw/workspace
gateway.mode: local
gateway.port: 18789
gateway.bind: lan

Config handling
Use existing values

QuickStart
Keeping your current gateway settings:
Gateway port: 18789
Gateway bind: LAN
Gateway auth: Token (default)
Tailscale exposure: Off
Direct to chat channels.

Model/auth provider
Gateway port: 18789
Gateway bind: LAN
Gateway auth: Token (default)
Tailscale exposure: Off
Direct to chat channels.

Model/auth provider
  OpenAI
  Anthropic
  Chutes
  vLLM
  MiniMax
  Moonshot AI (Kimi K2.5)
  Google
  xAI (Grok)
  Mistral AI
  Volcano Engine
  BytePlus
  OpenRouter (API key)
  Kilo Gateway
  Qwen
  Z.AI
  Qianfan
  Copilot
  Vercel AI Gateway
  OpenCode Zen
  Xiaomi
  Synthetic
  Together AI
  Hugging Face
  Venice AI
  LiteLLM
  Cloudflare AI Gateway
  Custom Provider
  Skip for now
```

Next, at this step, we need to select a model. Due to OpenRouter's traffic control strategy, we recommend that you reserve more than \$11 in your OpenRouter account. For insufficient balance, OpenRouter may rate limit (this policy may be modified later, please refer to the latest actual situation). Here we can choose to use a free model (model name ends with free), or choose an OpenRouter free dedicated model openrouter/free (this is a router that will randomly select a free model from the models available on OpenRouter. This router will intelligently filter out models that support the functions required by your request)

```
◆ Default model
...
○ openrouter/anthropic/claude-haiku-4.5
○ openrouter/anthropic/claude-opus-4
○ openrouter/anthropic/claude-opus-4.1
○ openrouter/anthropic/claude-opus-4.5
○ openrouter/anthropic/claude-opus-4.6
○ openrouter/anthropic/claude-sonnet-4
○ openrouter/anthropic/claude-sonnet-4.5
○ openrouter/anthropic/claude-sonnet-4.6
○ openrouter/arcee-ai/trinity-large-preview:free
○ openrouter/arcee-ai/trinity-mini
○ openrouter/arcee-ai/trinity-mini:free
○ openrouter/arcee-ai/virtuoso-large
○ openrouter/openrouter/auto
○ openrouter/baidu/ernie-4.5-21b-a3b
○ openrouter/baidu/ernie-4.5-vl-28b-a3b
○ openrouter/bytedance-seed/seed-1.6
○ openrouter/bytedance-seed/seed-1.6-flash
○ openrouter/bytedance-seed/seed-2.0-mini
○ openrouter/cohere/command-r-08-2024
○ openrouter/cohere/command-r-plus-08-2024
○ openrouter/deepseek/deepseek-chat
○ openrouter/deepseek/deepseek-chat-v3-0324
○ openrouter/deepseek/deepseek-chat-v3.1
○ openrouter/deepseek/deepseek-v3.1-terminus
○ openrouter/deepseek/deepseek-v3.1-terminus:exacto
○ openrouter/deepseek/deepseek-v3.2
○ openrouter/deepseek/deepseek-v3.2-exp
○ openrouter/deepseek/deepseek-r1
○ openrouter/deepseek/deepseek-r1-0528
○ openrouter/essentialai/anj-1-instruct
● openrouter/openrouter/free (Free Models Router · ctx 195k · reasoning)
○ openrouter/google/gemini-2.0-flash-90k
...

```

At this time, an option requiring API Key may appear. We paste the API Key and press Enter to proceed to the next step

Later we come to other configuration items. Let's skip these for now and select Skip all

```
Zalo: Vietnam-focused messaging platform with Bot API.
Zalo Personal: Zalo personal account via QR code login.
Synology Chat: Connect your Synology NAS Chat to OpenClaw with full agent capabilities.
Tlon: decentralized messaging on Orbit; install the plugin to enable.

◆ Select channel (QuickStart)
  Skip for now
Commit overwrite: /root/.openclaw/openclaw.json (sha256 46b602c1773a9e88f0b137cdbcddb6696097642bd99b7bc2d503af8622fb4704 -> 63f2e0cf73860460b5329f95f22d6f322f4e6f0ee6c4622c2224eddfae07bd6f, backup=/root/.openclaw/openclaw.json.bak)
Updated ~/.openclaw/openclaw.json
Workspace OK: ~/.openclaw/workspace
Sessions OK: ~/.openclaw/agents/main/sessions

◆ Web search
  Web search lets your agent look things up online.
  Choose a provider and paste your API key.
  Docs: https://docs.openclaw.ai/tools/web

◆ Search provider
  Skip for now

◆ Skills status
  Eligible: 4
  Missing requirements: 40
  Unsupported on this OS: 7
  Blocked by allowlist: 0

◆ Configure skills now? (recommended)
  ○ Yes / ● No

```

```
Running agents on your computer is risky – harden your setup:
https://docs.openclaw.ai/security

◇ Dashboard ready
Dashboard link (with token):
http://192.168.2.56:18789/#token=7814bc0c9670673701379ea20e0296fe53eb91cf6d18a3a4
Copy/paste this URL in a browser on this machine to control OpenClaw.
No GUI detected. Open from your computer:
ssh -N -L 18789:127.0.0.1:18789 user@<host>
Then open:
http://localhost:18789/
http://localhost:18789/#token=7814bc0c9670673701379ea20e0296fe53eb91cf6d18a3a4
Docs:
https://docs.openclaw.ai/gateway/remote
https://docs.openclaw.ai/web/control-ui

◇ Web search
Web search was skipped. You can enable it later:
openclaw configure --section web
Docs: https://docs.openclaw.ai/tools/web

◇ What now
What now: https://openclaw.ai/showcase ("What People Are Building").

Onboarding complete. Use the dashboard link above to control OpenClaw.
root@jetson-yahboom:/#
```

At this point, it means we have completed the configuration

5.2 Modify Configuration File (Recommended Method)

Configuration File Location

The configuration file is located at: `/root/.openclaw/openclaw.json`

Edit Configuration File

1. Open Configuration File

```
nano /root/.openclaw/openclaw.json
# Or use your favorite editor, such as vim, code, etc.
```

2. Add or Modify Model Provider Configuration

Add or modify the `models.providers` section in the configuration file:

```
{
  "models": {
    "providers": {
      "openai": {
        "apiKey": "OPENAI_API_KEY"
      },
      "anthropic": {
        "apiKey": "ANTHROPIC_API_KEY"
      },
      "google": {
        "apiKey": "GOOGLE_API_KEY"
      }
    }
  },
}
```

```

"agents": {
  "defaults": {
    "model": {
      "primary": "openai/gpt-4"
    }
  }
}

```

3. Configuration Description

- `models.providers`: Configure API-KEY for each model service provider
 - `openai.apikey`: OpenAI's API-KEY
 - `anthropic.apikey`: Anthropic's API-KEY
 - `google.apikey`: Google's API-KEY
- `agents.defaults.model.primary`: Set the default model to use
 - Format is `service provider/model name`, such as `openai/gpt-4`, `anthropic/claude-3`, `google/gemini-pro`

4. Save and Exit

- When modifying, be careful not to randomly delete periods or add other symbols
- Press `Ctrl+O` to save, `Ctrl+X` to exit (nano editor)

5. Factory Default Bailian Configuration (Can be directly modified or referenced)

```

"models": {
  "mode": "merge",
  "providers": {
    "bailian": {
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
      "apiKey": "sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxx",
      "api": "openai-completions",
      "models": [
        {
          "id": "qwen3.5-plus",
          "name": "qwen3.5-plus",
          "reasoning": false,
          "input": [
            "text",
            "image"
          ],
          "contextwindow": 1000000,
          "maxTokens": 65536,
          "cost": {
            "input": 0,
            "output": 0,
            "cacheRead": 0,
            "cachewrite": 0
          }
        }
      ],
    },
  },
}

```



```

    "id": "MiniMax-M2.5",
    "name": "MiniMax-M2.5",
    "reasoning": false,
    "input": [
      "text"
    ],
    "cost": {
      "input": 0,
      "output": 0,
      "cacheRead": 0,
      "cachewrite": 0
    },
    "contextwindow": 196608,
    "maxTokens": 32768
  },
  {
    "id": "glm-4.7",
    "name": "glm-4.7",
    "reasoning": false,
    "input": [
      "text"
    ],
    "cost": {
      "input": 0,
      "output": 0,
      "cacheRead": 0,
      "cachewrite": 0
    },
    "contextwindow": 202752,
    "maxTokens": 16384,
    "compat": {
      "thinkingFormat": "qwen"
    }
  }
]
}

```

6. Common Problems and Solutions

6.1 API-KEY Invalid or Error

Problem: After configuration, it shows API-KEY invalid or authentication failed

Solution:

1. Confirm the API-KEY is correct (note no extra spaces)
2. Check if the API-KEY has expired or been disabled
3. Confirm if account balance is sufficient (some service providers require prepaid balance)

6.2 Cannot Access Model Service

Problem: Configuration is correct but cannot connect to model service

Solution:

1. Check if network connection is normal
2. Confirm if your region's network can access the service provider (some services may require special network configuration)
3. Try changing to another model service provider for testing